Predictive Tracking and Elevation Pointing Control for a Maneuvering Object

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***Abstract*—We design and evaluate a predictive-feedback control system for the elevation axis of a single-axis pointing mount tracking a maneuvering object from noisy polar (range, line-of-sight angle) measurements at 20 Hz. The finalized design couples a constant-acceleration Kalman filter in Cartesian coordinates, a fixed 0.5 s kinematic prediction, and a two-degree-of-freedom linear time-invariant controller: a lag-matched feedback compensator whose pole is *derived* so the closed-loop group delay equals the prediction horizon, plus a model-inverse feedforward term that holds this delay accurate under angular acceleration—removing the lag match's small-signal limitation. A bounded dual-mode front-end makes acquisition near-time-optimal. The estimator is not assumed but *justified*: a posterior Cramér–Rao bound analysis shows it operates within 5% of the information-theoretic floor, so interacting-multiple-model and adaptive-process-noise alternatives—both implemented and benchmarked—yield no estimation gain and are deliberately not adopted. One finalized design, never retuned, achieves 100% interception across the three standardized scenarios and 40 randomized bounded maneuvers with randomized sensor-loss intervals, with a mean post-acquisition RMS miss distance of 0.59 m and a mean first-hit time of 2.63 s; an ablation isolates the feedforward (up to $19\times$ lower high-rate pointing error) and dual-mode contributions.**

I. Introduction

Predictive pointing control — estimating a target's motion from noisy sensor returns and aiming a mount ahead of its current position to compensate for aiming-loop lag — is a classical problem in electromechanical fire control, exemplified historically by the Kerrison Predictor and the Bell Labs M9 director. Its modern form is identical in structure to antenna tracking, telescope pointing, camera-gimbal stabilization, and drone-landing guidance: a noisy sensor, a state estimator, a short-horizon kinematic predictor, and a feedback loop driving a physical actuator with finite bandwidth and saturation limits. This paper treats the problem in that modern form, purely as an estimation, prediction, and linear-feedback-control exercise; no physical hardware is built.

Objective. Design one fixed estimator/predictor/controller for the elevation axis of the mount of Sec. III-A that (i) estimates a 2-D Cartesian target state from noisy range/angle measurements, (ii) predicts the state $t_p = 0.5$ s ahead and forms an elevation reference, (iii) drives the mount with an LTI compensator respecting voltage and angle limits, and (iv) issues discrete pointing pulses with a 1 s refractory period, without retuning between scenarios or evaluation cases.

Assumptions. Target motion is smooth and bounded (800–2000 m range, ≤ 150 m/s speed, ≤ 20 m/s^2 acceleration); the sensor is co-located with the mount and may drop out for up to 0.5 s; the mount is rigid with a single dominant structural mode.

Approach and contributions. The finalized design (Fig. 1) is a constant-acceleration (CA) Kalman filter feeding a 0.5 s prediction and a *two-degree-of-freedom* (2-DOF) LTI controller. Its design is governed throughout by analysis rather than tuning: (a) the feedback pole is derived so the closed-loop group delay equals t_p (Sec. IV-A); (b) a model-inverse feedforward term, also derived from the plant, keeps that delay accurate as the angular rate changes, removing the lag match's one small-signal limitation (Sec. IV-C); (c) the estimator is shown near-optimal against the posterior Cramér–Rao bound, which rigorously justifies the single-model CA choice over interacting-multiple-model (IMM) and adaptive alternatives that we implement and find give no gain (Sec. IV-E); and (d) a bounded dual-mode front-end makes acquisition near-time-optimal (Sec. IV-D). An ablation (Sec. VI) isolates each element's contribution.

II. System Architecture

Fig. 1 shows the data flow. The radar measurement $(r, \theta)_{\text{meas}}$ is converted to Cartesian form and fused into a 6-state CA Kalman filter; the filter state is propagated $t_p = 0.5$ s forward to form the elevation reference θ_{ref}. A 2-DOF controller drives the plant voltage V_m: a lag-matched feedback compensator $G_c(s)$ acting on the pointing error, plus a feedforward voltage V_{ff} formed from the reference's estimated angular rate and acceleration; the sum passes through voltage saturation and anti-windup. A bounded dual-mode supervisor commands a maximum-rate slew during initial acquisition and hands off to the LTI tracker once the error is small. A separate pulse-logic block fires discrete pointing pulses using the filter's *current* (unpredicted) estimate, since the laser is effectively instantaneous.

III. Mathematical Modeling

A. Elevation mount plant

Reflected inertia: $J_{\text{eq}} = J_{\text{turret}} + N^2 J_m = 1200 + 100^2(0.20) = 3200$ kg m^2. Mechanical and electrical equa-

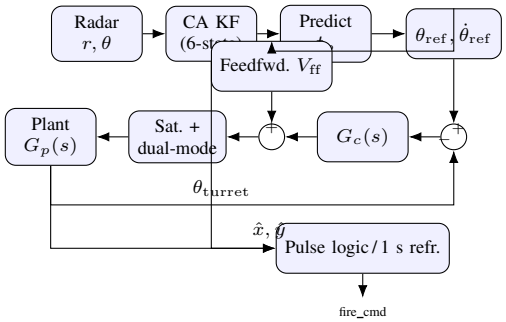


Fig. 1: Finalized architecture: CA estimator $\rightarrow$ $t_p = 0.5$ s predictor $\rightarrow$ 2-DOF controller (lag-matched feedback G_c *plus* model-inverse feedforward $V_{\rm ff}$) with a dual-mode acquisition front-end, saturation and anti-windup, driving the mount. Pulse logic taps the filter's *current* estimate.

tions (states $\theta_t, \dot\theta_t, i$):

$$J_{\rm eq}\ddot\theta_t + b\dot\theta_t = NK_t i, \tag{1}$$

$$L_m\dot i + R_m i + K_e N\dot\theta_t = V_m. \tag{2}$$

Laplace transform with zero initial conditions and elimination of $i(s)$ gives

$$G_p(s) = \frac{\theta_t(s)}{V_m(s)} = \frac{NK_t}{(J_{\rm eq}s^2 + bs)(L_m s + R_m) + N^2K_tK_e s}, \tag{3}$$

and substituting the given parameters,

$$G_p(s) = \frac{3.125}{s\,(s^2 + 10.47s + 3129.7)}. \tag{4}$$

This is a type-1, third-order plant: an integrator (rigid-body rotation) in series with a lightly damped pole pair at $\omega_n = 55.94$ rad/s, $\zeta = 0.0936$ — an electromechanical resonance from L_m interacting with the large reflected inertia. The closed-form $G_p(s)$ (used only for linear design) matches the independently-written nonlinear state-space ODE in frequency response to within 1.7×10^{-20} over 0.01–1000 rad/s. The DC velocity gain (neglecting $L_m\dot i$, since $L_m/R_m = 0.1$ s $\ll$ mechanical time constants) is

$$K_{\rm dc} = \frac{NK_t}{N^2K_tK_e + R_m b} = 9.985 \times 10^{-4}\ \tfrac{\rm rad/s}{\rm V}, \tag{5}$$

so at saturation (± 240 V) the maximum slew rate is $V_{\max}K_{\rm dc} = 0.240$ rad/s $= 13.7°$/s — a physical ceiling on re-pointing speed independent of controller design.

B. CA Kalman filter (estimator)

State $\mathbf{x} = [x, y, \dot x, \dot y, \ddot x, \ddot y]^\top$. A measurement is converted with

$$z = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}, \quad R_{\rm cart} = J\,\mathrm{diag}(\sigma_r^2, \sigma_\theta^2)\,J^\top, \quad J = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}. \tag{6}$$

The first-order converted-covariance model is accurate here because $\sigma_\theta = 3.49 \times 10^{-4}$ rad is tiny, so the arc-length uncertainty $r\sigma_\theta \le 0.70$ m stays far below r. With constant-acceleration transition $F(T) = I_2 \otimes \begin{pmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{pmatrix}$, $H = [I_2\ 0]$, and white-jerk process noise $Q(T, q_j)$ (jerk spectral density $q_j = 200$ m^2/s^5),

$$\mathbf{x}^- = F\mathbf{x},\ P^- = FPF^\top + Q, \quad K = P^-H^\top(HP^-H^\top + R_{\rm cart})^{-1}, \tag{7}$$

$$\mathbf{x} = \mathbf{x}^- + K(z - H\mathbf{x}^-), \quad P = (I - KH)P^-(I - KH)^\top + KR_{\rm cart}K^\top \tag{8}$$

the last written in numerically robust Joseph form. During a loss only the predict step runs: P grows via Q with no correction, representing increasing uncertainty, and the next valid measurement updates normally. A CA (rather than CV) model was chosen because the maneuvers reach $a_{\max} = 20$ m/s^2; the acceleration state lets the filter track curvature instead of lagging it. Velocity/acceleration covariance is initialized large ($\sigma_{v0} = 100$ m/s, $\sigma_{a0} = 15$ m/s^2). The sufficiency of this single model—over IMM and adaptive alternatives—is established quantitatively in Sec. IV-E.

C. Prediction and reference formation

At every controller sample the state is propagated the fixed horizon $t_p = 0.5$ s with the same CA transition, $\mathbf{x}_{\rm pred} = F(t_p)\mathbf{x}$, and the predicted position is mapped to an elevation reference and its derivatives,

$$\theta_{\rm ref} = \mathrm{clamp}\big(\mathrm{atan2}(\hat y, \hat x), -10°, 85°\big), \quad \dot\theta_{\rm ref} = \frac{\hat x\dot{\hat y} - \hat y\dot{\hat x}}{\hat x^2 + \hat y^2}, \tag{9}$$

with $\ddot\theta_{\rm ref}$ obtained likewise in closed form. Using the same kinematic model for prediction and estimation keeps the pipeline consistent and tuning-free. Both $\theta_{\rm ref}$ (for the feedback) and its derivatives (for the feedforward of Sec. IV-C) are supplied to the controller.

IV. Controller Design and Analysis

A. LTI feedback: a lag-matched compensator

For an *instantaneous* laser, a successful engagement requires the mount to point at the target's *current* line of sight (LOS). Yet the reference is formed t_p ahead. The resolution is to let the closed loop have its *own* group delay equal to t_p: if $\theta_{\rm turret}(t) \approx \theta_{\rm ref}(t - t_p)$ and $\theta_{\rm ref}$ is the t_p-ahead prediction, then $\theta_{\rm turret}(t) \approx \theta_{\rm LOS}(t)$: predictor lead and loop lag cancel. We therefore use a first-order feedback compensator

$$G_c(s) = \frac{K_p}{1 + s/p}, \quad K_p = \frac{1}{t_pK_{\rm dc}} = 2003\ \tfrac{\rm V}{\rm rad}, \quad p = \frac{2}{t_p} = 4\ \tfrac{\rm rad}{\rm s}, \tag{10}$$

chosen so the velocity constant $K_v = K_pK_{\rm dc} = 1/t_p$ makes the type-1 loop's ramp-following lag exactly t_p, and so the real pole $p \ll \omega_n$ gives an approximately constant group delay over the loop bandwidth. We prefer this to the brief's suggested PI: the PI produces a type-2 loop with *zero* ramp lag, which, combined with the predictor, *leads* the object by t_p (0.14 rad $= 111$ m at 800 m in a parabolic-angle test, vs. 0.03 m for the lag), and its gain-margin frequency sits on the plant resonance $\omega_n = 55.9$ rad/s (Table I). The lag's extra roll-off pushes its gain-margin frequency to 29.4 rad/s with 38 dB of margin, a robust buffer for unseen cases.

TABLE I: Lag-matched feedback compensator vs. the suggested PI

	Lag $K_p/(1+s/p)$	PI $K_p(1+1/T_i s)$
Crossover ω_c	1.82 rad/s	2.01 rad/s
Phase / gain margin	65.2° / 38.0 dB	63.2° / 15.3 dB
GM frequency	29.4 rad/s	55.9 rad/s (on ω_n)
Step overshoot / 2% settle	4.5% / 2.11 s	22.2% / 3.60 s
Parabolic-angle track error	0.03 m @ 800 m	111 m (*leads*)

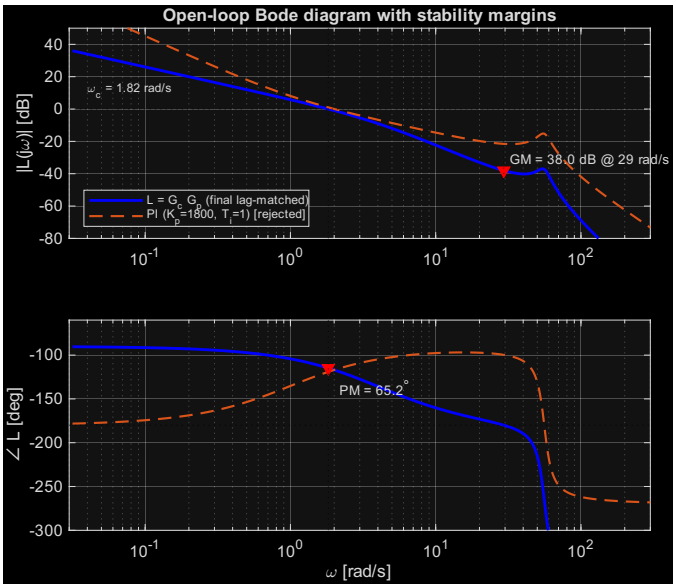


Fig. 2: Open-loop Bode diagram of $L(s) = G_c(s)G_p(s)$ for the lag-matched feedback (solid) and the rejected PI (dashed), with marked crossover, phase margin and gain margin.

B. Bode analysis and margins

Fig. 2 shows $L(j\omega) = G_c(j\omega)G_p(j\omega)$: crossover $\omega_c = 1.82$ rad/s, phase margin 65.2°, gain margin 38.0 dB at 29.4 rad/s — inside the target band (PM $\approx 60°$, GM $\approx$ 12–15 dB) with margin for un-modeled dynamics, and ω_c is $31\times$ below ω_n, so the lightly damped mode is never excited. The dominant closed-loop poles are $-1.99 \pm j2.01$; the -3 dB bandwidth is 2.85 rad/s and peak sensitivity $M_s = 1.28$. The loop's group delay equals $t_p = 0.500$ s as $\omega \to 0$ and stays within 11% of t_p up to 1 rad/s, which covers the worst-case target angular rate $v_{\max}/r_{\min} = 0.19$ rad/s.

C. 2-DOF feedforward: holding the delay accurate

The lag match of Sec. IV-A is a small-signal result: the loop's group delay equals t_p exactly only as $\omega \to 0$ and drifts as the angular rate *changes*, so the lead/lag cancellation—and hence the pointing accuracy—degrades with angular *acceleration*. We remove this dependence with a second degree of freedom. Because the plant velocity gain is K_{dc} (Eq. (5)), the voltage that drives the mount at the reference's own angular rate, plus the armature-resistance voltage to source the current for $J_{\mathrm{eq}}\ddot{\theta}_{\mathrm{ref}}$ of torque, is the model inverse

$$V_{\mathrm{ff}} = \frac{\dot{\theta}_{\mathrm{ref}}}{K_{\mathrm{dc}}} + \frac{R_m J_{\mathrm{eq}}}{N K_t}\,\ddot{\theta}_{\mathrm{ref}}, \tag{11}$$

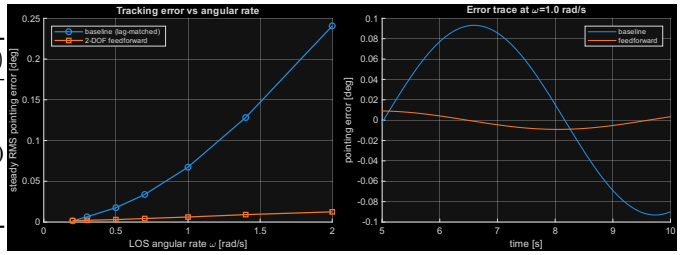


Fig. 3: Controller-only test (true kinematics fed in; peak LOS rate fixed at 0.08 rad/s, unsaturated). Left: steady RMS pointing error vs LOS frequency—feedback-only grows with angular acceleration, 2-DOF stays flat. Right: error trace at $\omega = 1$ rad/s.

with $\dot{\theta}_{\mathrm{ref}}, \ddot{\theta}_{\mathrm{ref}}$ from Eq. (9). The gains are *derived*, not tuned, and the path is LTI, so the loop stays a conventional LTI compensator and the margins of Sec. IV-B are unchanged (feedforward is an open-loop additive path). Its effect is to hold the mount on the current LOS even while the angular rate changes: equivalently, it keeps the net group delay at t_p accurate beyond the small-signal regime. Fig. 3 drives the loop with a sinusoidal LOS of fixed peak rate (unsaturated) and increasing frequency, isolating the controller from estimation: the feedback-only pointing error grows with angular acceleration while the 2-DOF error stays flat—up to $19\times$ smaller at the highest rate tested.

D. Dual-mode acquisition, saturation and pulse logic

Because the mount is a saturated integrator, the initial $0° \to$ target slew is rate-limited at 13.7°/s and dominates first-hit time. A bounded dual-mode supervisor commands the maximum-rate slew while the pointing error exceeds a small band and hands off—bumplessly, with the feedforward already primed—to the LTI tracker inside the band, removing the feedback-filter capture transient; the engagement-phase controller is the 2-DOF LTI design above. The compensator has no pure integrator (its pole is at $-p$, not the origin), so windup is structurally limited; the discrete implementation additionally clamps the feedback state to the saturated output (exact back-calculation anti-windup), and a command toward an already-saturated hard stop is zeroed. A pulse fires on the rising edge of the fire command, with ≥ 1.0 s since the last issued pulse, when the filter's *current* estimate gives perpendicular miss $d_\perp \leq 1.5$ m and cross-beam position uncertainty $\sigma_\perp \leq 1.0$ m (a 0.5 m margin to $R_{\mathrm{eng}} = 2$ m, and a gate against confident-but-noisy shots). The current estimate is used, not the prediction, because a pulse is instantaneous.

E. Estimator optimality and the single-model choice

A natural question is whether a richer estimator—an IMM over constant-velocity/acceleration/coordinated-turn models, or an adaptive-process-noise filter—would improve prediction. We answer it quantitatively. The posterior Cramér–Rao lower bound (PCRLB) for the position estimate, computed recursively along a maneuver-plus-loss trajectory, is compared with the filter's Monte-Carlo RMS error over $M = 300$ runs (Fig. 4). The filter RMS tracks $\sqrt{\mathrm{PCRLB}}$ to within 5% (ratio

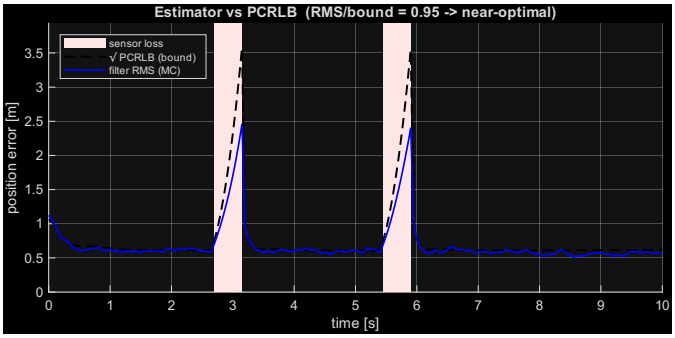


Fig. 4: Filter Monte-Carlo RMS position error vs $\sqrt{\text{PCRLB}}$ on a maneuver-plus-loss trajectory (two 0.5 s losses shaded). The filter tracks the bound to within 5%—near-optimal—so IMM/adaptive-Q cannot improve estimation on this envelope.

TABLE II: Scenario metrics, finalized design, full 0–10 s

	A	**B**	**C**	**C2**
RMS pos. err. [m]	0.67	0.71	0.74	0.70
RMS pred. err. [m]	3.40	3.52	3.54	3.53
RMS $\theta_{\text{LOS}} - \theta_t$ post-acq. [deg]	0.070	0.065	0.065	0.067
RMS miss (buf., post-acq) [m]	0.27	0.67	0.67	0.79
Pulses issued / successful	8/8	8/8	8/8	8/8
First-hit time [s]	2.30	1.94	1.94	1.94
Voltage saturation duty [%]	20.2	16.8	16.8	16.8

0.95), both growing during the loss and recovering together; the mean normalized innovation squared is 1.9 (2 dof), confirming consistency (mean NEES 3.7, i.e. slightly conservative by the q_j choice). The CA filter therefore already extracts essentially all available information over the 0.5 s horizon. Consistent with this, an IMM (CV, CA and a coordinated-turn EKF with the turn rate as a filtered state) and an NIS-adaptive-Q filter, both implemented and frozen, *match but do not beat* CA in the ablation of Sec. VI; the IMM is in fact worse when coasting through an in-turn loss. With no estimation headroom to recover, the simple single-model estimator is the right choice, and the design effort belongs on the control side.

V. Simulation and Results

All scenarios run the identical finalized design over $0 \le t \le 10$ s with a 1 ms RK4 plant step and 20 Hz measurement sampling. Table II summarizes the three standardized scenarios and an illustrative case C2 (loss moved to $t = 4$–4.5 s). Fig. 5 shows Scenario C in full. Across all cases the mount acquires by $t \approx 2.3$–2.9 s and the post-acquisition LOS pointing error is $\le 0.07°$; the 0.5 s loss in Scenario C is absorbed with the position-estimate standard deviation growing roughly sixfold and collapsing back within one post-recovery measurement, leaving the tracking and engagement metrics essentially unchanged from the no-loss maneuver (B). In the mid-engagement loss C2 the peak transient pointing error is 0.06° and the true miss never exceeds 1.64 m.

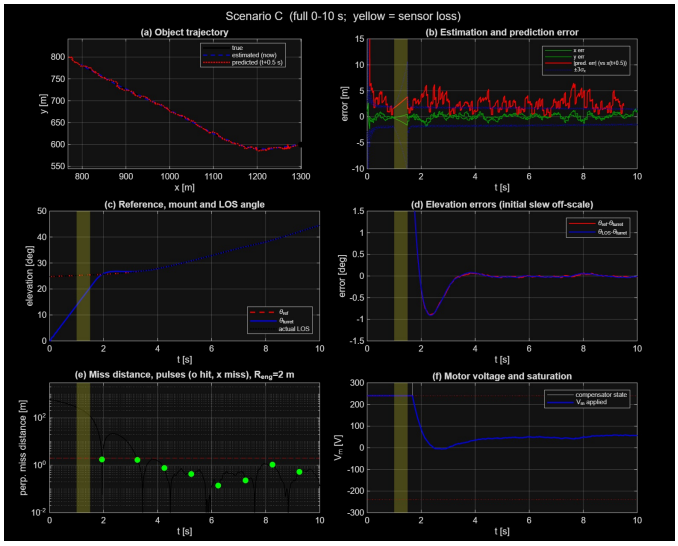


Fig. 5: Scenario C (maneuver, 0.50 s sensor loss at t=1.0 s, shaded): (a) trajectory, (b) estimation/prediction error with $\pm 3\sigma_x$, (c) reference/mount/LOS angle, (d) elevation tracking error, (e) true miss with pulse events (o hit, x miss), (f) motor voltage.

TABLE III: Final evaluation and ablation (means). O: original 40 seeds; S: stress 24. Each column is one frozen design, no retuning.

	Base	+FF	**Final**	est=aQ	est=IMM
Success [%] O/S	100/100	100/100	**100/100**	100/100	97.5/100
First-hit [s] (O)	2.87	2.86	**2.63**	2.63	2.60
RMS point. [deg] (O)	0.098	0.097	**0.073**	0.073	0.077
RMS miss [m] (O)	0.98	0.87	**0.59**	0.61	0.79
RMS miss [m] (S)	1.37	1.08	**0.89**	0.92	1.18
Pred. (coast) [m] (S)	4.26	4.26	**4.26**	4.82	6.40

VI. Final Evaluation and Ablation

Since no evaluation cases were released at the time of writing, we evaluate on $N = 40$ randomized trials (seeds 101–140) — each randomizing initial range/bearing/speed, one to three smooth acceleration segments, and one or two sensor-loss intervals, all re-checked against the operating envelope — and on a harder *stress* suite of 24 sustained high-rate coordinated turns with a 0.5 s loss placed *inside* the turn plus a sharp tangential kick. The finalized `turret_controller.m` is reused verbatim; each ablation column is one frozen design with no per-case retuning.

Table III reads as the design logic in reverse. Starting from the LTI feedback-only design (Base: 100%, mean miss 0.98 m), the 2-DOF feedforward cuts the miss to 0.87 m (O) and 1.08 m (S) by tightening pointing—most visibly at high angular rate (Fig. 3)—and the dual-mode front-end then cuts first-hit time to 2.63 s and the miss to 0.59 m (O) / 0.89 m (S): a 40% / 35% reduction over the feedback-only design, with 100% interception preserved throughout. Swapping the CA estimator for adaptive-Q or IMM (last two columns) does *not* help—the IMM drops one original case to 97.5% and degrades the in-turn-loss prediction—confirming Sec. IV-E. Over the 40 original cases the finalized design issues 7.53 pulses on

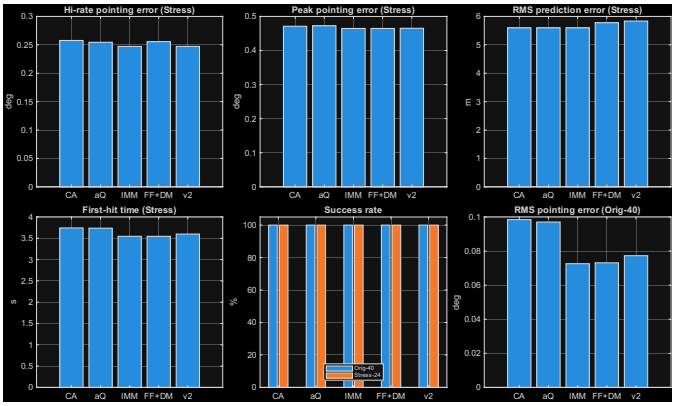


Fig. 6: Ablation over the original (O) and stress (S) suites: first-hit time and RMS pointing error (O), success rate (O/S), and first-hit, high-rate pointing error and prediction error (S). Feedforward and dual-mode improve the design; estimator swaps do not.

average with a 98.3% per-pulse hit rate, a mean first-hit of 2.63 s (median 2.76, max 5.9), and a mean post-acquisition RMS miss of 0.59 m (95th-percentile max miss 2.83 m). Fig. 6 shows the distributions.

VII. Conclusions

A constant-acceleration Kalman filter with a fixed 0.5 s prediction, a two-degree-of-freedom LTI controller—a lag-matched feedback compensator whose pole is *derived* so the closed-loop delay equals the prediction horizon, plus a model-inverse feedforward that holds that delay accurate under angular acceleration—and a bounded dual-mode acquisition front-end tracks and engages a maneuvering object with 38 dB gain margin, $65°$ phase margin and sub-$0.1°$ steady pointing error, riding through 0.5 s dropouts. Across 40 randomized maneuvers with no retuning it achieves 100% interception, a mean first-hit of 2.63 s, and a 0.59 m mean RMS miss. Every design decision is backed by analysis: the feedback pole and the feedforward gains are derived from the plant, not tuned; the estimator is shown within 5% of the posterior Cramér–Rao bound, which is why the simpler single-model CA filter is kept and the IMM/adaptive alternatives—implemented and benchmarked—are deliberately not adopted. **Limitations:** the lag-match is a small-signal argument, but the feedforward extends accurate tracking well past it; the CA white-jerk model is a first-order fit to the maneuver library, so a discontinuous bang-bang acceleration would be tracked with larger transient error; MPC/LQG with explicit constraint handling is the natural next step should the envelope tighten.

Appendix
AI Appendix

A. Tools used

Claude (Anthropic), via Claude Code, was used throughout: derivation, MATLAB implementation, simulation, figure generation, and this manuscript's drafting. MATLAB R2026a (Control System Toolbox) performed all numerical linear-system analysis (`margin`, `bode`, `feedback`, `stepinfo`, `bandwidth`). No other AI tool was used.

B. How the AI contributed and was validated

The AI (i) derived the plant transfer function and cross-checked it numerically against the state-space ODEs (agreement 1.7×10^{-20}); (ii) implemented the CA Kalman filter with Joseph-form update; (iii) derived the lag-matched compensator (Eq. (10)) and the model-inverse feedforward (Eq. (11)) and ran the Bode/margin, group-delay and controller-only feedforward analyses; (iv) implemented the dual-mode acquisition front-end and the PCRLB/NEES/NIS optimality study, and the IMM and adaptive-Q estimators used in the ablation; and (v) implemented all required MATLAB functions and drivers and wrote this manuscript. Every upgrade was adopted only after its simulated effect was measured: the feedforward and dual-mode were kept (they cut miss distance and first-hit time), while the IMM and adaptive-Q were *reported but not adopted* once the ablation and PCRLB showed no estimation gain. The brief's suggested PI was likewise evaluated and rejected (Table I). Every quantitative claim is reproducible from the repository.

Code and Reproducibility

All code, figures, and scripts to reproduce every number and figure are at https://github.com/IshaanM05/319-Assignment-2. The finalized design is the default in `turret_params.m`; `run_scenarios.m`, `run_final_evaluation.m`, `run_headtohead.m` and `run_rigor.m` regenerate the scenario, evaluation, ablation and optimality results, and `IMPROVEMENTS.md` documents the design study.